Facile Synthesis of C_3-Symmetric Tripodal β-Hydroxy Amide Ligands and Their Ti (IV) Complex-Catalyzed Asymmetric Addition of Diethylzinc to Aldehydes

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Abstract: Chiral C_3-symmetric tripodal tris(β-hydroxy amide) ligands have been conveniently synthesized *via* the reaction of 1,3,5-benzenetricarboxylic chloride and chiral amino alcohols, and the corresponding tris(amino alcohol) ligands were obtained *via* the borane reduction of amides. The asymmetric diethylzinc addition to various aldehydes catalyzed by these chiral ligand−Ti (IV) complexes gave modest enantioselectivities.

Keywords: C_3-symmetry, β-hydroxy amide, amino alcohol, asymmetric synthesis, diethylzinc.

INTRODUCTION

C_2-Symmetry is an important structural feature in many chiral metal catalysts since it can reduce the number of possible diastereomeric intermediates or transition states [1]. C_2-Symmetric bidentate ligands have been successfully applied in the field of asymmetric synthesis [1,2]. C_3-Symmetry is also intriguing in chemistry and many C_3-symmetric compounds and their applications in chiral molecular recognition and asymmetric catalysis have been reported. Recent reviews have illustrated the importance of C_3-symmetry in asymmetric catalysis and chiral recognition [3,4]. C_3-Symmetric chiral ligands in octahedral complexes permit a reduction in the number of possible diastereomorphous transition states, as for C_2-symmetric systems in tetrahedral or square-planar environments. The development of new types of C_3-symmetic ligands for asymmetric reaction has attracted attention during recent years [4]. For example, Chan and co-workers have reported that the C_3-symmetric oxazolinyl ligand catalyzed the addition of diethylzinc to aromatic aldehydes to give secondary alcohols with high enantiomeric excesses (up to 90%) [5]. Katsuki and co-workers have developed tridentate trisoxazoline ligands as chiral auxiliary in asymmetric allylic oxidation of cycloalkenes and moderate to excellent enantioselectivities (up to 93% ee) were obtained [6]. Ahn's group has reported interesting results of C_3-symmetric trisoxazolines in the asymmetric catalysis and the molecular recognition of aminonium ions and sugars [7]. Gade and co-workers also reported that chiral C_3-symmetric trisoxazolines are highly efficient stereo-controlling ligands in the Cu(II) catalyzed enantioselective α-amination (up to 99% ee) as well as the enantioselective Mannich reaction of prochiral β-ketoesters (up to 91% ee) [8]. Bringmann reported the enantioselective synthesis of a novel-type C_3-symmetric tripodal ligand that is composed of a central mesitylene-derived core and three functionalized, axially chiral biaryl subunits. The triol is a suitable catalyst for the enantioselective addition of dialkylzinc to various aromatic aldehydes with asymmetric induction of up to 98% ee [9]. However, to the best of our knowledge, the use of C_3-symmetric hydroxy amide ligands in diethylzinc addition has not been reported so far. It is interesting to investigate further the catalytic ability of β-hydroxy amide derivatives. We expected that the cooperation of amido group and hydroxy group in these ligands may result in special catalytic activity. In our previous reports, C_3-symmetirc β-hydroxy amide ligands have been demonstrated as excellent catalysts for the asymmetric alknylation of aldehydes and the borane reduction of ketones [10]. Herein, the synthesis and applications of C_3-symmetric tripodal tris(β-hydroxy amide) and tris(amino alcohol) ligands in the asymmetric diethylzinc addition of aldehydes are reported.

For carbon-carbon bond-forming reactions, the catalytic asymmetric addition of diethylzinc to aldehydes is a versatile method for synthesizing optically active secondary alcohols [11]. Over the past years, the asymmetric addition of organozinc reagents to aldehydes in the presence of the catalytic amount of chiral ligands have attracted considerable attention and many ligands inducing high enantioselectivity have been reported [12]. Although many significant results have been achieved in this area, much effort to develop new types of cheap and efficient chiral catalysts for this important asymmetric reaction are still in great need. Herein, we provide an alternative ligand design approach for the asymmetric diethylzinc addition.

RESULTS AND DISCUSSION

A series of C_3-symmetric tris(β-hydroxy amide) ligands were conveniently prepared from commercially available or easily synthesized optically pure amino alcohols in excellent yields. A mixture of 1,3,5-benzenetricarboxylic acid, $SOCl_2$

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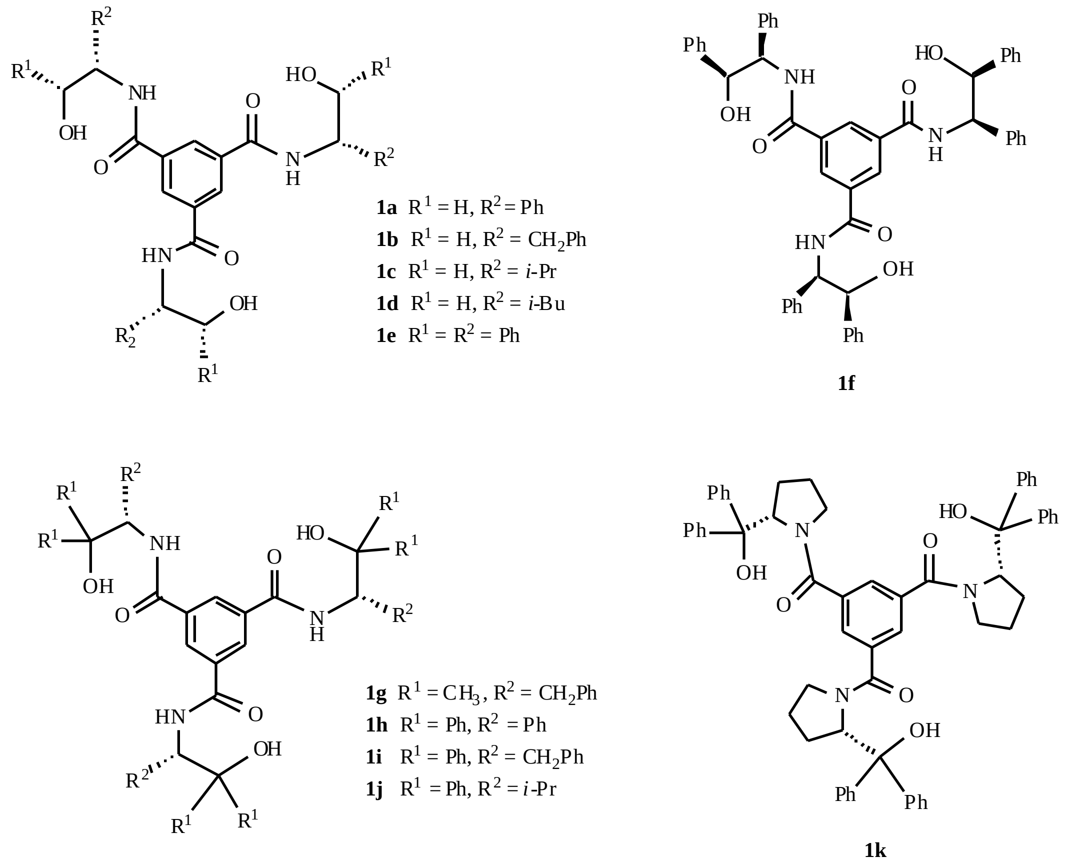
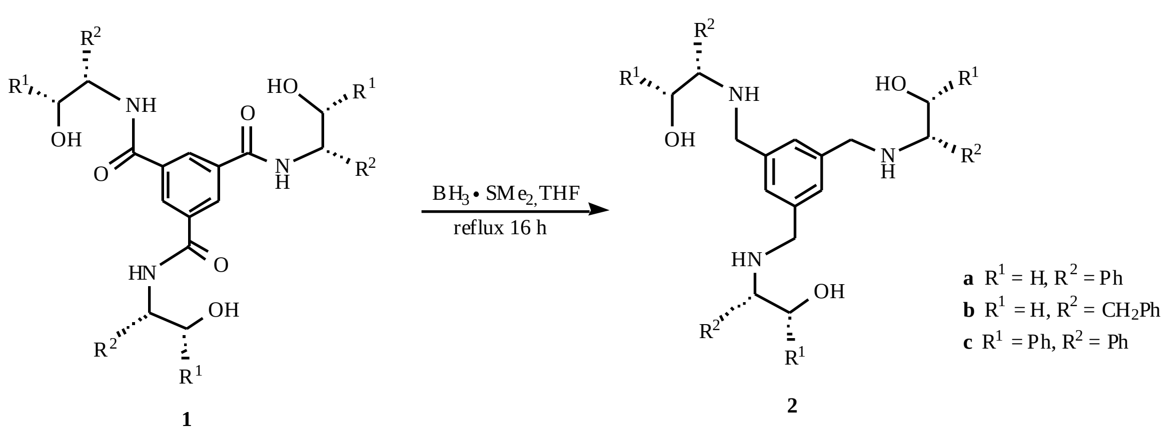


Fig. (1). C_3-symmetric tris(β-hydroxy amide) ligands.

and several drops of DMF was refluxed to afford 1,3,5-benzenetricarboxylic chloride. After removal of excess $SOCl_2$ and without further purification, the trichloride reacted with chiral amino alcohols [13] in the presence of excess triethylamine in CH_2Cl_2 at reflux to afford tris(β-hydroxy amide) **1a−1k** in 70-96% yields [10]. The structures of these ligands are shown in Fig. (**1**). Armstrong and co-workers have attempted the synthesis of homochiral C_3-symmetric amide **1b**, but they did not provide the analytical data owing to the difficulties in purifying this compound [14].

The C_3-symmetric tris(amino alcohol) ligands **2a**, **2b**, and **2c** can be easily obtained in 76−93% yields by the borane reduction of the corresponding C_3-symmetric tris(β-hydroxy amide) ligands for 16 h (Scheme **1**).

The asymmetric addition of diethylzinc to benzaldehyde using catalytic system prepared in situ by mixing $Ti(O^iPr)_4$ with a catalytic amount of chiral ligand was chosen as model reaction. First, after careful screen of solvent, ligand **1** (20 mol %) and $Ti(O^iPr)_4$ (1.4 equiv.) were used to catalyze the addition of diethylzinc (3 equiv.) to benzaldehyde in hexane−CH_2Cl_2 (1:2, V/V) at room temperature. The results are summarized in Table **1**. Ligand **1f** shows more effectively for benzaldehyde and gave the corresponding secondary alcohol with 47% ee in almost quantitative yield within 24 h. Other C_3-symmetric tris(β-hydroxy amide)



Scheme 1. Synthesis of C_3-symmetric tris(amino alcohol) ligands.

ligands and C_3-symmetric tris(amino alcohol) ligands **2a**, **2b**, and **2c** all gave lower enantioselectivities. Ligand **1e** is the best performing one in the asymmetric alkynylation [10a] and ligand **1f** is the most effective ligand in the asymmetric diethylzinc addition. This result indicates that the synergic effect of substituents at positions 1 and 2 of chiral amino alcohols is very important. Both of the enantiomeric isomers **1e** and **1f** have two adjacent *syn*-substituted phenyl groups, the chiral shielding effect may be the best one in these two asymmetric catalytic systems. Double substitution in position 1 of amino alcohols (ligands **1g**-**1k**) is detrimental to enantioselectivity, this may be ascribed to the disappearing of one chiral center near the hydroxy group. Interestingly, a slight increase ligand/Ti(O^{i}Pr)$_4$ ratio (from 1:7 to 1:5) improved the catalytic activity of **1f** (entry 16 vs 6). Further increase or decrease the ligand/Ti(O^{i}Pr)$_4$ ratio resulted in the decrease of enantioselectivies (entries 20−22). There is no serious loss of enantioselectivity when the reaction was performed at higher temperature (40 °C) (Table **1**, entry 15). The enantioselectivity was enhanced slightly (56% ee) at −20 °C, but lower temperature resulted in the decrease of product yields (entries 17−19).

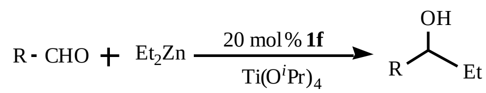
The asymmetric additions of diethylzinc to various aldehydes were conducted using the best catalytic ligand **1f**-Ti(O^{i}Pr)$_4$ system under the optimized reaction conditions and the results are summarized in Table **2**. The chiral secondary alcohols were obtained in good isolated yields of 71–80% and modest enantioselectivities of 50–69% ee for all aromatic aldehydes (Table **2**, entries 1–10). Aromatic aldehydes with electron-donating or electron-withdrawing groups show same level of enantioselectivities, and the electronic nature of the substitutes seems not to affect the enantioselectivity. *Ortho*-substituted aldehydes gave slightly lower enantioselectivities owing to steric effect. The diethylzinc addition to aliphatic aldehyde such as phenylacetaldehyde gave the highest enantioselectivity (70% ee) (Table **2**, entry 11). The addition reaction to α,β-unsaturated aldehyde such as *trans*-cinnamaldehyde was also examined and the product was obtained in the same level of enantioselectivity as other aldehydes (68% ee) (Table **2**, entry 12). The absolute configurations of the products were determined to be *R* for all aldehydes examined in this study by the comparison with the literature data [15-18].

Table 1. Enantioselective Diethylzinc Addition to Benzaldehyde[a]

Entry	Ligand	Ligand/Ti(O^{i}Pr)$_4$	T (°C)	Time (h)	Yield (%)[b]	ee (%)[c]	Config[d]
1	**1a**	1:7	rt	24	90	24	*S*
2	**1b**	1:7	rt	24	95	10	*S*
3	**1c**	1:7	rt	24	94	14	*R*
4	**1d**	1:7	rt	24	93	4	*R*
5	**1e**	1:7	rt	24	84	15	*S*
6	**1f**	1:7	rt	14	99	47	*R*
7	**1g**	1:7	rt	24	99	30	*S*
8	**1h**	1:7	rt	24	84	13	*S*
9	**1i**	1:7	rt	24	92	12	*R*
10	**1j**	1:7	rt	24	84	11	*R*
11	**1k**	1:7	rt	24	98	13	*R*
12	**2a**	1:7	rt	24	80	6	*R*
13	**2b**	1:7	rt	24	90	7	*R*
14	**2c**	1:7	rt	24	96	16	*S*
15	**1f**	1:5	40	14	99	42	*R*
16	**1f**	1:5	rt	14	96	52	*R*
17	**1f**	1:5	0	24	84	47	*R*
18	**1f**	1:5	−20	48	78[e]	56	*R*
19	**1f**	1:5	−40	20	83	49	*R*
20	**1f**	1:1	rt	14	92	2	*R*
21	**1f**	1:3	rt	14	99	40	*R*
22	**1f**	1:10	rt	14	99	37	*R*

[a]Reaction conditions: molar ratio of aldehyde:ligand:Et_2Zn = 1:0.2:3, hexane:CH_2Cl_2 =1:2 as solvent. [b]Determined by HPLC analysis. [c]Determined by HPLC analysis on Daicel Chiracel OB column. [d]The absolute configuration assigned by the comparison with the literature [15]. [e]Isolated yield after column chromatography.

Table 2. Enantioselective Diethylzinc Addition to Aldehydes[a]



Entry	R	Yield (%)[b]	Ee (%)[c]	Config[d]
1	C_6H_5	78	56	*R*
2	2-$CH_3C_6H_4$	76	56	*R*
3	4-$CH_3C_6H_4$	74	62	*R*
4	4-$MeOC_6H_4$	79	62	*R*
5	2-FC_6H_4	80	62	*R*
6	4-FC_6H_4	75	69	*R*
7	2-ClC_6H_4	78	50	*R*
8	4-ClC_6H_4	78	68	*R*
9	3-BrC_6H_4	74	62	*R*
10	1-Naphthyl	71	68	*R*
11	$C_6H_4CH_2$	76	70	*R*
12	*trans*-$C_6H_5CH{=}CH$	70	68	*R*

[a]Reaction conditions: molar ratio of aldehyde:Et_2Zn:$Ti(O^iPr)_4$ = 1:3:1, hexane:CH_2Cl_2 =1:2 as solvent at −20 °C. [b]Isolated yield after column chromatography. [c]Determined by HPLC analysis on a Daicel Chiracel OB, OD, or OD-H column. [d]The absolute configuration assigned by the comparison with the literature [15-18].

In summary, a small library of new C_3-symmetric tris(β-hydroxy amide) ligands **1** have been conveniently synthesized from easily available chiral amino alcohols in two steps with excellent yields. The corresponding C_3-symmetric tris(amino alcohol) ligands **2** were easily obtained by the borane reduction of the tris(β-hydroxy amide) ligands. C_3-symmetric tris(β-hydroxy amide) ligands **1** and tris(amino alcohol) ligands **2** were used to catalyze the diethylzinc addition to aldehydes and modest enantio-selectivities were obtained with ligand **1f**. Generally, ligands **1** are better than ligands **2** in the asymmetric catalysis. These results show that the C_3-symmetric tris(β-hydroxy amide) ligands have potential for asymmetric catalysis. Further study is ongoing in our laboratory to expand the application of these new chiral ligands to other asymmetric reactions.

EXPERIMENTAL

Melting points were measured on an XT-4 melting point apparatus and are uncorrected. NMR spectra were recorded on Varian 200 MHz, Varian 300 MHz, or Brucker 400 MHz spectrometer. Infrared spectra were obtained on a Bruker Vector 22 spectrometer or Nicolet AVATAR 330 FT-IR spectrometer. Mass spectra were obtained on a VG-ZAB-HS (EI) mass spectrometer or APEX II FT−ICRMS (FAB) mass spectrometer. Flash column chromatography was run over 200−300 mesh silica gel. The optical rotations were measured on a Perkin–Elmer 341 LC spectrometer. The enantiomeric excesses of addition products were determined by analytical high performance liquid chromatography (HPLC) on Aglilent HP1100 liquid chromatograph equipped with a DAD detector using the Diacel Chiralcel OB, OD, or OD-H column. HPLC grade isopropyl alcohol and *n*-hexane were used as eluting solvents. Toluene, diethyl ether and tetrahydrofuran were distilled with sodium under nitrogen prior to use. Methylene chloride was distilled with calcium. Other solvents used were purified and dried by standard procedures. All of the commercially available ketones were used as received without additional purification. Diethylzinc (1.0 M solution in hexane) was purchased from Acros. All reactions involving moisture-sensitive reagents were performed under a nitrogen atmosphere.

Synthesis of tris(β-hydroxy amide) 1a-1k

Ligands **1a-1g, 1i-1k** were synthesized in 70-96% yields according to our previous reported procedure [10]. The synthesis and characterization of new ligand **1h** was given in following.

N,N',N''-Tris[(1S)-2-hydroxy-1,2,2-triphenylethyl]-1,3,5-benzenetricarboxamide (1h)

A 25 mL flask was charged with 1,3,5-benzenetricar-boxylic acid (0.63 g, 3 mmol), $SOCl_2$ (7.5 g, 63 mmol) and several drops of DMF. The mixture was heated at reflux for 6 h forming a clear solution. Complete removal of excess $SOCl_2$ in vacuo and addition of CH_2Cl_2 (30 mL) provided a 0.1 M solution of 1,3,5-benzenetricarboxylic chloride. The solution of 1,3,5-benzenetricarboxylic chloride (2.8 mL, 0.1 M solution in CH_2Cl_2, 0.28 mmol) was added slowly to a solution of (*S*)-2-amino-1,1,2-triphenylethanol (0.25 g, 0.86 mmol) and Et_3N (0.5 mL) in CH_2Cl_2 (10 mL) at 0 °C. The mixture was stirred at reflux for 6 h. The reaction mixture was cooled to room temperature, and washed with water (2 × 5 mL). The organic layer was dried over anhydrous Na_2SO_4 and concentrated in vacuo to give the crude product. The crude product was purified by chromatography on silica gel (eluting with dichloromethane-methanol 15:1) to afford **1h** (230 mg, 80% yield) as colorless solid, mp 191−193 °C; $[\alpha]_D^{20}$ = −299.5 (*c* 1.49, AcOEt). IR (KBr): ν 3404, 3061, 3031, 1655, 1509, 1496, 1448, 1061, 910, 750, 697 cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): 7.87 (s, 3H), 7.54 (d, *J* = 7.6 Hz, 6H), 7.42 (d, *J* = 7.8 Hz, 3H), 7.26−7.04 (m, 36H), 6.10 (d, *J* = 8.6 Hz, 3H), 3.26 (d, *J* = 8.2 Hz, 3H); ^{13}C NMR (50 MHz, $CDCl_3$): δ 165.1, 144.2, 144.0, 137.5, 134.7, 128.6, 128.5, 128.2, 127.8, 127.7, 127.4, 126.9, 125.9, 125.6, 80.8, 59.7. HRMS (FAB): *m*/*z* calcd for $C_{69}H_{57}N_3O_6Na$: 1046.4139; Found: 1046.4174 $[M+Na^+]$.

Typical Procedure for the Reduction of Amide 1

A 25 mL round bottom flask was charged with 0.568 g (1.0 mmol) of triamide **1a** in THF (5 mL), 13.5 mL of BH_3 SMe_2 (27 mmol, 2 M in THF) was added. The reaction mixture was heated under reflux for 16 h. After cooling to room temperature, saturated NH_4Cl solution (4 mL) was added to destroy the excess of BH_3. After complete removal of organic solvent, 20% KOH aqueous solution (20 mL) was added and the mixture was refluxed for 6 h. The mixture was cooled and extracted with CH_2Cl_2 (3 × 20 mL). The organic phase was dried over anhydrous sodium sulfate. After concentration by rotatory evaporation, the product was purified by column chromatography on silica gel to afford **2a** 0.49 g (93% yield) as a colorless oil.

N,N',N''-Tris[(1S)-2-hydroxy-1-phenylethyl]-1,3,5-triaminomethylbenzene (2a)

93% yield; colorless oil; $[\alpha]_D^{20}$ = +64.7 (*c* 0.38, $CHCl_3$). IR (KBr): ν 3377, 2924, 1605, 1452, 1361, 1200, 1055, 759, 702 cm^{-1}. ^{1}H NMR (200 MHz, $CDCl_3$): δ 7.28−7.21 (m, 15H), 7.01 (s, 3H), 3.76−3.40 (m, 15H), 3.22 (br, 6H). ^{13}C NMR (50 MHz, $CDCl_3$): δ 140.33, 140.29, 128.5, 127.5, 127.4, 126.8, 66.6, 63.9, 50.9. HRMS (FAB): *m*/*z* calcd for $C_{33}H_{40}N_3O_3$: 526.3064; Found: 526.3070 $[M+H^+]$.

N,N',N''-Tris[(1S)-1-hydroxymethyl-2-phenylethyl]-1,3,5-triaminomethylbenzene (2b)

76% yield; colorless solid, mp132−134 ºC; $[\alpha]_D^{20}$ = +14.7 (*c* 0.034, C_2H_5OH). IR (KBr): ν 3292, 2923, 1604, 1495, 1454, 1111, 1059, 860, 746, 698 cm^{-1}. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.27-7.14 (m, 15H), 6.96 (s, 3H), 4.57 (s, 3H), 3.70-3.64 (m, 6H), 3.33-3.24 (m, 6H), 2.73-2.63 (m, 9H), 1.78 (dd, *J* = 6.8, 11.5 Hz, 3H); ^{13}C NMR (50 MHz, $CDCl_3$): δ140.7, 139.8, 129.3, 128.1, 125.8, 125.6, 62.3, 60.1, 50.6. HRMS (FAB): *m*/*z* calcd for $C_{36}H_{46}N_3O_3$: 568.3547; Found: 568.3544 $[M+H^+]$.

N,N',N''-Tris[(1S,2R)-2-hydroxy-1,2-diphenylethyl]-1,3,5-triaminomethylbenzene (2c)

82% yield; colorless solid, mp70−72 ºC; $[\alpha]_D^{20}$ = +8 (*c* 0.60, C_2H_5OH). IR (KBr): ν 3419, 3028, 2853, 1609, 1495, 1453, 1386, 1197, 1028, 759, 700 cm^{-1}. ^{1}H NMR (200 MHz, $CDCl_3$): δ 7.25−6.87 (m, 33H), 4.79 (d, *J* = 5.8 Hz, 1H), 4.57 (d, *J* = 8.4 Hz, 2H), 3.86 (d, *J* = 6.0 Hz, 2H), 3.68−3.42 (m, 8H), 2.97 (br, 6H). ^{13}C NMR (50 MHz, $CDCl_3$): δ 141.2, 140.1, 139.6, 128.4, 127.9, 127.7, 127.5, 127.32, 127.25, 126.6, 77.7, 69.6, 51.0. HRMS (FAB): *m*/*z* calcd for $C_{51}H_{52}N_3O_3$: 754.4003; Found: 754.3988 $[M+H^+]$.

Typical Procedure for the Asymmetric Addition of Diethylzinc to Aldehyde

Under nitrogen, to a solution of **1f** (39.8 mg, 0.05 mmol) in dichloromethane (1 mL) was added titanium tetraisopropoxide (72 μL, 0.25 mmol) and stirred for 1 h. Diethylzinc (0.75 mL, 0.75 mmol, 1.0 M in hexane) was added and the mixture was stirred for 0.5 h. The aldehyde (0.25 mmol) in dichloromethane (1 mL) was added at −20 ºC and the mixture was stirred for additional 48 h. The reaction was quenched with saturated ammonium chloride solution (2 mL), extracted with dichloromethane (3 × 5 mL) and dried over anhydrous sodium sulfate. After removal of the solvent under reduced pressure, the residue was purified by column chromatography to afford the pure chiral alcohol.

(R)-1-Phenyl-1-propanol

Colorless oil, $[\alpha]_D^{20}$ = +25.8 (*c* 1.0, $CHCl_3$). 56% ee determined by HPLC analysis (Daicel Chiralcel OB column, hexane: 2-propanol 97.5:2.5, 0.5 mL/min, 254 nm). Retention time: $t_{R,major}$ = 18.32 and $t_{R,minor}$ = 22.14 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.35−7.21 (m, 5H), 4.51 (t, *J* = 6.3 Hz, 1H), 2.45 (s, 1H), 1.82−1.64 (m, 2H), 0.87 (t, *J* = 7.2 Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 144.5, 128.2, 127.3, 125.9, 75.8, 31.7, 10.0. lit.[15] $[\alpha]_D^{25}$ = +42.4 (*c* 2.5, $CHCl_3$, 99% ee).

(R)-1-(2-Methylphenyl)-1-propanol

Colorless oil, $[\alpha]_D^{20}$ = +41.8 (*c* 1.2, $CHCl_3$). 56% ee determined by HPLC analysis (Daicel Chiralcel OD column, hexane: 2-propanol 95:5, 0.5 mL/min, 254 nm). Retention time: $t_{R,major}$ = 13.74 and $t_{R,minor}$ = 11.77 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.40 (d, *J* = 7.2 Hz, 1H), 7.21−7.08 (m, 3H), 4.78 (t, *J* = 6.4 Hz, 1H), 2.33 (s, 4H), 1.74−1.64 (m, 2H), 0.93 (t, *J* = 7.5 Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 142.7, 134.4, 130.1, 126.9, 126.0, 125.1, 71.8, 30.8, 18.9, 10.2. lit.[15] $[\alpha]_D^{25}$ = +64.10 (*c* 2.12, C_6H_6, 96% ee).

(R)-1-(4-Methylphenyl)-1-propanol

Colorless oil, $[\alpha]_D^{20}$ = +26.0 (*c* 0.7, $CHCl_3$). 62% ee determined by HPLC analysis (Daicel Chiralcel OB column, hexane: 2-propanol 95:5, 0.5 mL/min, 254 nm). Retention time: $t_{R,major}$ = 16.22 and $t_{R,minor}$ = 14.48 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.20−7.10 (m, 4H), 4.46 (t, *J* = 6.9 Hz, 1H), 2.52 (s, 1H), 2.31(s, 3H), 1.80−1.61 (m, 2H), 0.85 (t, *J* = 7.5 Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 141.6, 136.8, 128.9, 125.8, 75.6, 31.6, 20.9, 10.0. lit.[15] $[\alpha]_D^{25}$ = +41.62 (*c* 2.53, C_6H_6, 99% ee).

(R)-1-(4-Methoxyphenyl)-1-propanol

Colorless oil, $[\alpha]_D^{20}$ = +21.6 (*c* 2.30, CH_2Cl_2). 62% ee determined by HPLC analysis (Daicel Chiralcel OD-H column, hexane: 2-propanol 90:10, 1.0 mL/min, 220 nm). Retention time: $t_{R,major}$ = 12.10 and $t_{R,minor}$ = 11.15 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.22 (d, *J* = 8.7Hz , 2H), 6.85 (d, *J* = 8.7Hz , 2H), 4.48 (t, *J* = 6.6Hz, 1H), 3.77 (s, 1H), 2.33 (brs, 1H), 1.83−1.63 (m, 2H), 0.86 (t, *J* = 7.5Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 158.8, 136.7, 127.1, 113.6, 75.4, 55.1, 31.6, 10.1. lit.[15] $[\alpha]_D^{25}$ = +30.41 (*c* 3.57, C_6H_6, 91% ee).

(R)-1-(2-Fluorophenyl)-1-propanol

Colorless oil, $[\alpha]_D^{20}$ = +11.2 (*c* 0.8, $CHCl_3$). 62% ee determined by HPLC analysis (Daicel Chiralcel OB column, hexane: 2-propanol 95:5, 0.5 mL/min, 254 nm). Retention time: $t_{R,major}$ = 13.59 and $t_{R,minor}$ = 11.50 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.46−7.40 (m, 1H), 7.26−7.10 (m, 2H), 7.03−6.96 (m, 1H), 4.91 (t, *J* = 6.6 Hz, 1H), 2.41 (s, 1H), 1.83−1.72 (m, 2H), 0.91 (t, *J* = 7.2 Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 159.8 (*J* = 243.9Hz), 131.5 (*J* = 7.0Hz), 128.6 (*J* = 8.3Hz), 127.2 (*J* = 4.8Hz), 124.1 (*J* = 3.5Hz), 115.2 (*J* = 22.0Hz), 69.5, 30.9, 9.8. lit.[16] $[\alpha]_D^{25}$ = +32.6 (*c* 0.633, $CHCl_3$, 97% ee).

(R)-1-(4-Fluorophenyl)-1-propanol [17]

Colorless oil, $[\alpha]_D^{20}$ = +28.3 (*c* 1.55, CH_2Cl_2). 69% ee determined by HPLC analysis (Daicel Chiralcel OB column, hexane: 2-propanol 97:3, 0.8 mL/min, 220 nm). Retention time: $t_{R,major}$ = 20.39 and $t_{R,minor}$ = 19.37 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.21−7.16 (m, 2H), 6.98−6.90 (m, 2H), 4.43 (t, *J* = 6.6Hz, 1H), 2.79 (brs, 1H), 1.73−1,54 (m, 2H), 0.79 (t, *J* = 7.5Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 162.0 (*J* = 243.3Hz), 140.2 (*J* = 3Hz), 127.5 (*J* = 8.3Hz), 115.0 (*J* = 20.8Hz), 75.1, 31.8, 9.9.

(R)-1-(2-Chlorophenyl)-1-propanol

Colorless oil, $[\alpha]_D^{20}$ = +3.1 (*c* 1.2, $CHCl_3$). 50% ee determined by HPLC analysis (Daicel Chiralcel OB column, hexane: 2-propanol 95:5, 0.5 mL/min, 254 nm). Retention time: $t_{R,major}$ = 15.57 and $t_{R,minor}$ = 12.04 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.49 (dd, *J* = 1.8, 7.8 Hz, 1H), 7.31−7.13 (m, 3H), 5.01 (t, *J* = 5.7 Hz, 1H), 2.81 (s, H), 1.82−1.64 (m, 2H), 0.94 (t, *J* = 7.5 Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 141.9, 131.8, 129.2, 128.2, 127.1, 126.9, 71.7, 30.4, 9.9. lit.[15] $[\alpha]_D^{25}$ = +52.31 (*c* 3.46, $CHCl_3$, 96% ee).

(R)-1-(4-Chlorophenyl)-1-propanol

Colorless oil, $[\alpha]_D^{20}$ = +24.2 (*c* 2.60, CH_2Cl_2). 68% ee determined by HPLC analysis (Daicel Chiralcel OD-H column, hexane: 2-propanol=95:5, 0.5 mL/min, 220 nm). Retention time: $t_{R,major}$ = 31.22 and $t_{R,minor}$ = 29.62 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.34−7.16 (m, 4H), 4.49 (t, *J* = 6.6Hz, 1H), 2.61 (brs, 1H), 1.78−1.60 (m, 2H), 0.83 (t, *J* = 7.5Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 142.9, 132.9, 128.3, 127.3, 75.1, 31.8, 9.9. lit.[15] $[\alpha]_D^{25}$ = +27.32 (*c* 2.91, C_6H_6, 98% ee).

(R)-1-(3-Bromophenyl)-1-propanol [12r]

Colorless oil, $[\alpha]_D^{20}$ = +22.6 (*c* 3.65, CH_2Cl_2). 62% ee determined by HPLC analysis (Daicel Chiralcel OD-H column, hexane: 2-propanol 95:5, 1.0 mL/min, 220 nm). Retention time: $t_{R,major}$ = 15.12 and $t_{R,minor}$ = 13.79 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.50 (s, 1H), 7.42−7.38 (m, 2H), 7.27−7.20 (m, 2H), 4.58−4.53 (m, 1H), 2.0 (brs, 1H), 1.83−1.67 (m, 2H), 0.91 (t, *J* = 7.5Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 146.8, 130.3, 129.8, 128.9, 124.5, 122.3, 75.0, 31.7, 9.8.

(R)-1-(1-Naphthyl)-1-propanol

Colorless oil, $[\alpha]_D^{20}$ = +41.2 (*c*1.2, $CHCl_3$). 68% ee determined by HPLC analysis (Daicel Chiralcel OB column, hexane: 2-propanol 95:5, 0.5 mL/min, 254 nm). Retention time: $t_{R,major}$ = 21.73 and $t_{R,minor}$ = 19.66 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 8.02−7.35 (m, 7H), 5.23 (dd, *J* = 5.4, 7.2 Hz, 1H), 2.55 (s, 1H), 1.95−1.74 (m, 2H), 0.92 (t, *J* = 7.5 Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 140.1, 133.6, 130.4, 128.7, 127.6, 125.3, 123.1, 122.8, 72.2, 30.9, 10.4. lit.[15] $[\alpha]_D^{25}$ = +51.10 (*c* 4.10, $CHCl_3$, 98% ee).

(R)-1-phenylbutan-2-ol

Colorless oil, $[\alpha]_D^{20}$ = -19.4 (*c* 2.45, CH_2Cl_2). 70% ee determined by HPLC analysis (Daicel Chiralcel OD-H column, hexane: 2-propanol 95:5, 1.0 mL/min, 220 nm). Retention time: $t_{R,major}$ = 12.89 and $t_{R,minor}$ = 9.75 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.36−7.18 (m, 4H), 3.74−3.65 (m, 1H), 2.78 (dd, *J* = 4.5, 13.5Hz, 1H), 2.61 (dd, *J* = 8.4, 13.5Hz, 1H), 1.91 (brs, 1H), 1.57−1.41 (m, 2H), 0.94 (t, *J* = 7.5Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 138.6, 129.3, 128.4, 126.2, 73.9, 43.4, 29.4, 9.9. lit.[18] $[\alpha]_D^{28}$ = −20.2 (*c* 7.16, $CHCl_3$, 73% ee).

(R)-E-1-Phenyl-1-penten-3-ol

Colorless oil, $[\alpha]_D^{20}$ = +4.5 (*c* 1.3, $CHCl_3$). 68% ee determined by HPLC analysis (Daicel Chiralcel OB column, hexane: 2-propanol 95:5, 0.5 mL/min, 254 nm). Retention time: $t_{R,major}$ = 16.80 and $t_{R,minor}$ = 17.78 min. ^{1}H NMR (300 MHz, $CDCl_3$): δ 7.40−7.19 (m, 5H), 6.54 (d, *J* = 16.2 Hz, 1H), 6.19 (dd, *J* = 6.6, 16.2 Hz, 1H), 4.20−4.13 (m, 1H), 2.43 (s, 1H), 1.69−1.57 (m, 2H), 0.94 (t, *J* = 7.5 Hz, 3H). ^{13}C NMR (75 MHz, $CDCl_3$): δ 136.7, 132.2, 130.2, 128.4, 127.4, 126.3, 74.2, 30.1, 9.7. lit.[15] $[\alpha]_D^{25}$ = +14.74 (*c* 3.46, $CHCl_3$, 96% ee).

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